PRE-ALGEBRA (8TH) · REAL NUMBERS AND EXPONENTS

Square Roots on the Number Line

Explore & Reflect

Name _____ Date _____ Period _____

Open **Square Roots on the Number Line** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Between which two whole numbers is √40? How do you know without a calculator?

2. Is √50 equal to 25? Explain.

2 Explore use the tool

1. Choose area 40 and zoom in once. Between which tenths is √40?

2. Zoom in again. Between which hundredths is √40?

3. Choose area 49 and zoom in. What is different?

4. Type estimates for √20 until the square is within 0.05 of 20. Record your best estimate and its square.

3 Record use the tool and fill in the table

Trap the square root

Area	Between whole numbers	Between tenths	Rational or irrational?
30			
64			
10			
75			
2			

4 Explain what did you discover?

1. √A is the _____ of a square whose area is A.

2. To trap $\sqrt{A}$, find two numbers whose _____ are just below and just above A.

3. The square root of a whole number that is not a perfect square is _____: its decimal never ends and never _____.

4. Zooming showed $\sqrt{2} \approx 1.414\ldots$. Does zooming prove that $\sqrt{2}$ is irrational? Explain.

5 Apply a new situation, no tool

1. Order from least to greatest: $\sqrt{30}$, 5.4, $\frac{11}{2}$, $\sqrt{36}$.

2. Estimate $\sqrt{90}$ to the nearest tenth without a calculator.

3. A square garden has an area of 200 ft^2. About how long is each side?

4. Which of these are irrational: $\sqrt{16}$, $\sqrt{17}$, $\frac{1}{3}$, $\sqrt{0.25}$?